

BUSINESS ANALYTICS PROJECT REPORT

Dynamic Contextual Pricing Engine with Multi-Armed Bandits (Thompson Sampling)

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Week	Week 2
Subject	Business Analytics
Task Name	Dynamic Contextual Pricing Engine with Multi-Armed Bandits (Thompson Sampling)

Task Description

Students will design a dynamic pricing simulation. The algorithm must represent prices as continuous arms of a bandit, calculate contextual posterior Beta distributions based on real-time consumer conversion ratios, execute Thompson Sampling to balance exploration-exploitation trade-offs, and dynamically adjust prices in response to simulated competitive shocks.

1. Project Objective

The objective is to develop a dynamic pricing simulation in which candidate prices are treated as arms of a multi-armed bandit. The pricing engine observes consumer conversion behavior, updates contextual posterior distributions, selects prices through Thompson Sampling, and adapts pricing decisions when competitive conditions change.

The system is designed as a simulation. The task description does not prescribe a specific dataset, programming language, number of price arms, or numerical shock configuration, so these should remain configurable.

2. Dynamic Pricing and Context

The pricing engine receives contextual information that may influence conversion behavior. Example contexts can include demand conditions, customer segment, time period, traffic source, or a simulated competitor price. The exact context variables should be defined according to the implementation.

Example contextual input:

```
context = {  
    demand_level,  
    customer_segment,  
    competitor_price,  
    time_period  
}
```

candidate prices:
 $P = \{p_1, p_2, \dots, p_k\}$

At each round:

1. Observe current context.
2. Select a price.
3. Observe conversion outcome.
4. Update posterior parameters.
5. Select the next price.

3. Continuous Price Arms

The task requires prices to be represented as continuous arms. Instead of treating pricing as a fixed collection of unrelated categorical choices, the simulation can define a continuous price interval and evaluate candidate prices sampled from that range.

Price range:

$p_{\min} \leq p \leq p_{\max}$

Example:

$p_{\min} = 80$

$p_{\max} = 120$

Candidate prices may be generated or refined within this continuous interval.

Price_t = selected price at round t

The precise price range and discretization or continuous optimization strategy are not specified by the task description. The implementation should document the selected configuration.

4. Contextual Conversion Model

Each selected price produces a consumer response in the simulation. A conversion is represented as a binary outcome: 1 for conversion and 0 for no conversion. The observed conversion ratio is used to update the Beta posterior.

For price/context arm a :

conversions _{a} = number of successful conversions
failures _{a} = number of non-conversions

observed conversion rate:

$CR_a = \text{conversions}_a / (\text{conversions}_a + \text{failures}_a)$

Binary observation:

$\text{reward}_t \in \{0, 1\}$

This allows the engine to continuously incorporate new consumer behavior into its posterior estimates.

5. Beta Posterior Distribution

The Beta distribution provides a convenient posterior model for a Bernoulli conversion probability. For each contextual pricing arm, prior parameters are updated after observing conversion and non-conversion outcomes.

```
Prior:  
 $\theta_a \sim \text{Beta}(\alpha_a, \beta_a)$   
  
After a conversion:  
 $\alpha_a \leftarrow \alpha_a + 1$   
  
After a non-conversion:  
 $\beta_a \leftarrow \beta_a + 1$   
  
Posterior:  
 $\theta_a \mid \text{data} \sim \text{Beta}(\alpha_a, \beta_a)$   
  
Posterior mean:  
 $E[\theta_a] = \alpha_a / (\alpha_a + \beta_a)$ 
```

A contextual implementation can maintain separate posterior parameters for different context-price combinations or use a richer model to share information across contexts.

6. Thompson Sampling

Thompson Sampling balances exploration and exploitation by sampling a conversion probability from each arm's current posterior distribution and selecting the arm with the highest sampled value.

For each candidate price a :

$\theta_a \sim \text{Beta}(\alpha_a, \beta_a)$

Select:

$a^* = \text{argmax}_a \theta_a$

Observe conversion reward r .

Update:

if $r = 1$:

$\alpha_{a^*} \leftarrow \alpha_{a^*} + 1$

else:

$\beta_{a^*} \leftarrow \beta_{a^*} + 1$

The randomness in posterior sampling allows uncertain pricing options to be explored while increasingly favoring options supported by observed conversion behavior.

7. Revenue-Aware Pricing

Because the project concerns pricing, the simulation can evaluate revenue in addition to conversion probability. A simple realized revenue measure is price multiplied by the binary conversion outcome.

$$\text{Revenue}_t = \text{Price}_t \times \text{Conversion}_t$$

Expected revenue estimate:

$$E[\text{Revenue}_a] \approx \text{Price}_a \times E[\text{Conversion}_a]$$

$$= \text{Price}_a \times \alpha_a / (\alpha_a + \beta_a)$$

This distinction is important because the price with the highest conversion probability is not necessarily the price with the highest expected revenue. The exact reward definition should be stated in the project implementation.

8. Simulated Competitive Shocks

The simulation introduces competitive shocks that modify the market environment. For example, a simulated competitor price change can alter the conversion behavior associated with the available prices.

Example shock:

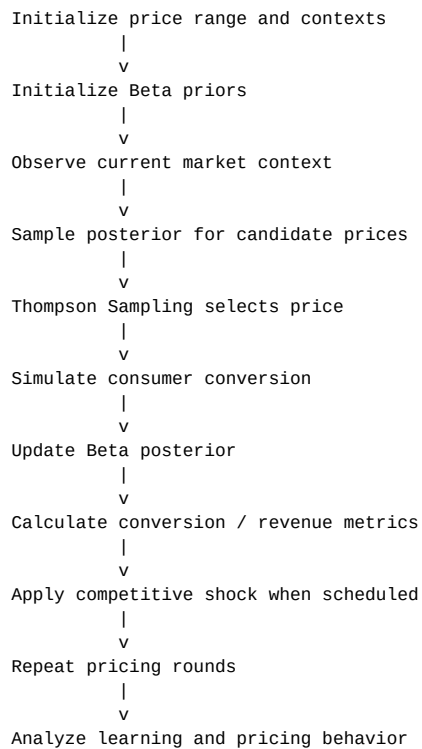
```
if competitor_price decreases:  
    reduce simulated conversion probability  
    for affected price levels
```

```
if competitor_price increases:  
    increase or restore simulated  
    conversion probability
```

```
After the shock:  
    observe new conversions  
    update Beta posteriors  
    continue Thompson Sampling
```

The task requires dynamic adjustment in response to simulated competitive shocks. The exact magnitude, timing, and direction of the shock are configuration choices and should be recorded in the experiment.

9. End-to-End Simulation Workflow



The simulation should log each round so that price decisions, conversion outcomes, posterior changes, competitive conditions, and revenue can be analyzed after execution.

10. Performance, Evaluation and Deliverables

Evaluation Metrics: conversion rate, revenue per round, cumulative revenue, selected price over time, posterior mean, exploration frequency, response after competitive shocks, and learning convergence.

Visualizations: price versus time, conversion rate versus time, cumulative revenue, posterior distributions for selected prices, and competitive shock markers.

Testing: verify Beta parameter updates, confirm Thompson Sampling selects among available price candidates, test repeated contexts, introduce competitive shocks at controlled rounds, and compare behavior before and after shocks.

Expected Deliverables:

1. Dynamic pricing simulation source code.
2. Continuous price-arm configuration.
3. Context and conversion simulation module.
4. Beta posterior update implementation.
5. Thompson Sampling decision engine.
6. Competitive shock simulation.
7. Pricing and revenue analytics.
8. Visualizations and experiment results.
9. Final project report.

Conclusion

This Week-2 Business Analytics task develops a dynamic contextual pricing engine using multi-armed bandits and Thompson Sampling. The simulation combines continuous pricing decisions, contextual conversion observations, Beta posterior updates, exploration-exploitation balancing, and adaptation to simulated competitive shocks.